

Test on patterns, sequences, and series

Marks: 60 Time: 1 hour 15 minutes

1. Write a program to print based on user's choice

$$6 \times 10 = 60$$

(1) $\begin{array}{c} * \\ * * \\ * * * \\ * * * * \\ * * * * * \end{array}$

(2) $\begin{array}{c} * * * * * \\ * * * \\ * \end{array}$

(3) $\begin{array}{c} * * * * * \\ * * * * * \\ * * * * * \\ * * * * * \end{array}$

(4) $\begin{array}{c} * \# * \# * \\ * \# * \# * \\ * \# * \# * \\ * \# * \# * \end{array}$

2. Write a program to print based on user's choice

(1) $\begin{array}{ccccc} * & * & * & * & * \\ & * & * & * & * \\ & & * & * & * \\ & & & * & * \\ & & & & * \end{array}$

(2) $\begin{array}{ccccc} & * & & & * \\ & & * & & * \\ & & & * & \\ & & & & * \\ * & & & & * \end{array}$

(3) $\begin{array}{cccccc} * & * & * & * & * & * \\ & * & & & & * \\ & & * & & & * \\ & & & * & & * \\ & & & & * & * \\ * & * & * & * & * & * \end{array}$

(4) $\begin{array}{cccccccc} & & & & & & & * \\ & & & & & & * & * \\ & & & & & * & * & * \\ & & & * & * & * & * & * \\ & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * \end{array}$

3. Write a program to print patterns based on user's choice

(1)	1 3 5 7 9	(2)	1 2 3 4 5	(3)	1 2 3 4 5	(4)	5 5 5 5 5
	3 5 7 9		1 2 3 4		1 2 3 4 4		4 4 4 4
	5 7 9		1 2 3		1 2 3 4 3		3 3 3
	7 9		1 2		1 2 4 3 2		2 2
	9		1		1 4 3 2 1		1

4. Write a program to print the Floyd's triangle and Reversed Floyd's triangle based on user's choice

(1)

1				
2	3			
4	5	6		
7	8	9	10	
11	12	13	14	15

(2)

15	14	13	12	11
10	9	8	7	
6	5	4		
3	2			
1				

5. Write a program to take choice from user and find S or print the sequence

(a) $S = 1 + \frac{x}{2} + \frac{x^2}{3} + \frac{x^3}{4} + \dots + \text{upto } n\text{th terms}$

(b) 0, 3, 8, 15, 24, 35 99

(c) 2, 5, 10, 17, 26, 37 101

6. Write a program to take choice from user and find S

(a) $S = 1 + (1 + 2) + (1 + 2 + 3) + (1 + 2 + 3 + 4) + \dots + (1 + 2 + 3 + 4 + 5 + \dots + n)$

$$(b) S = \frac{x}{2!} + \frac{x^2}{3!} + \frac{x^3}{4!} + \dots + \frac{x^n}{n!}$$

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